Image of 2? Preimages of 1?

5. Thread · MCQ On the globe, two points on the same parallel have:

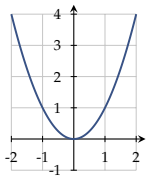
A. the same latitude **B.** the same longitude **C.** the same distance from Greenwich **D.** the same altitude

Variations and extrema of a function

Sheet 43 Thu 11 Feb. 2027 *Increasing, decreasing, monotonicity*

1. Warm-up · Quick $f(x) = -2x + 5$. Calculate $f(1)$ and $f(3)$; what do you notice?
2. Core · MCQ f is increasing on I means that, for all $a < b$ in I:

   A. $a \leqslant b$ only **B.** $f(a) \geqslant f(b)$ **C.** $f(a) \leqslant f(b)$ **D.** $f(a) = f(b)$
3. Core + · Graph Curve of $f(x) = x^2$ on $[-2;2]$:



   Give the variations of f.
4. Review · Quick Which multiplier corresponds to an increase of 15 %? To a decrease of 15 %?
5. Thread · Quick Is 457 divisible by 3? By 9? Justify.

Sheet 44 Mon 1st Mar. 2027 *Table of variations; maximum and minimum on an interval*

1. Warm-up · Quick Calculate x^2 for $x = -2, 0, 3$.
2. Core · Method f is decreasing on $[-3;1]$ with $f(-3) = 5$, $f(1) = -1$, then increasing on $[1;4]$ with $f(4) = 4$. Give the maximum and the minimum of f on $[-3;4]$.
3. Core + · True/false True or false? Justify. The maximum of a function on $[a;b]$ is always reached at a or at b.
4. Review · Table Is this table a proportionality table?

x	1	2	3
y	5	9	13

5. Thread · Method Is the triangle with sides 5, 12 and 13 right-angled? Which property should be used?

Sheet 45 Tue 2 Mar. 2027 *Affine functions: rate of change and variations*

1. Warm-up · Quick $f(x) = 3x - 1$. Calculate $\dfrac{f(4) - f(1)}{4 - 1}$.
2. Core · Quick Direction of variation of $f(x) = -4x + 7$?
3. Core + · Proof Show that $f(x) = 5x - 2$ is increasing on $\mathbb{R}$.
4. Review · Table Ages of the members of a club.

Age	10	11	12	13
Frequency	3	6	10	6

   How many members are at most 11 years old? What relative frequency is that?
5. Thread · Programme → formula Programme: multiply by 3 then subtract 2. Give $f(x)$ then $f(4)$.